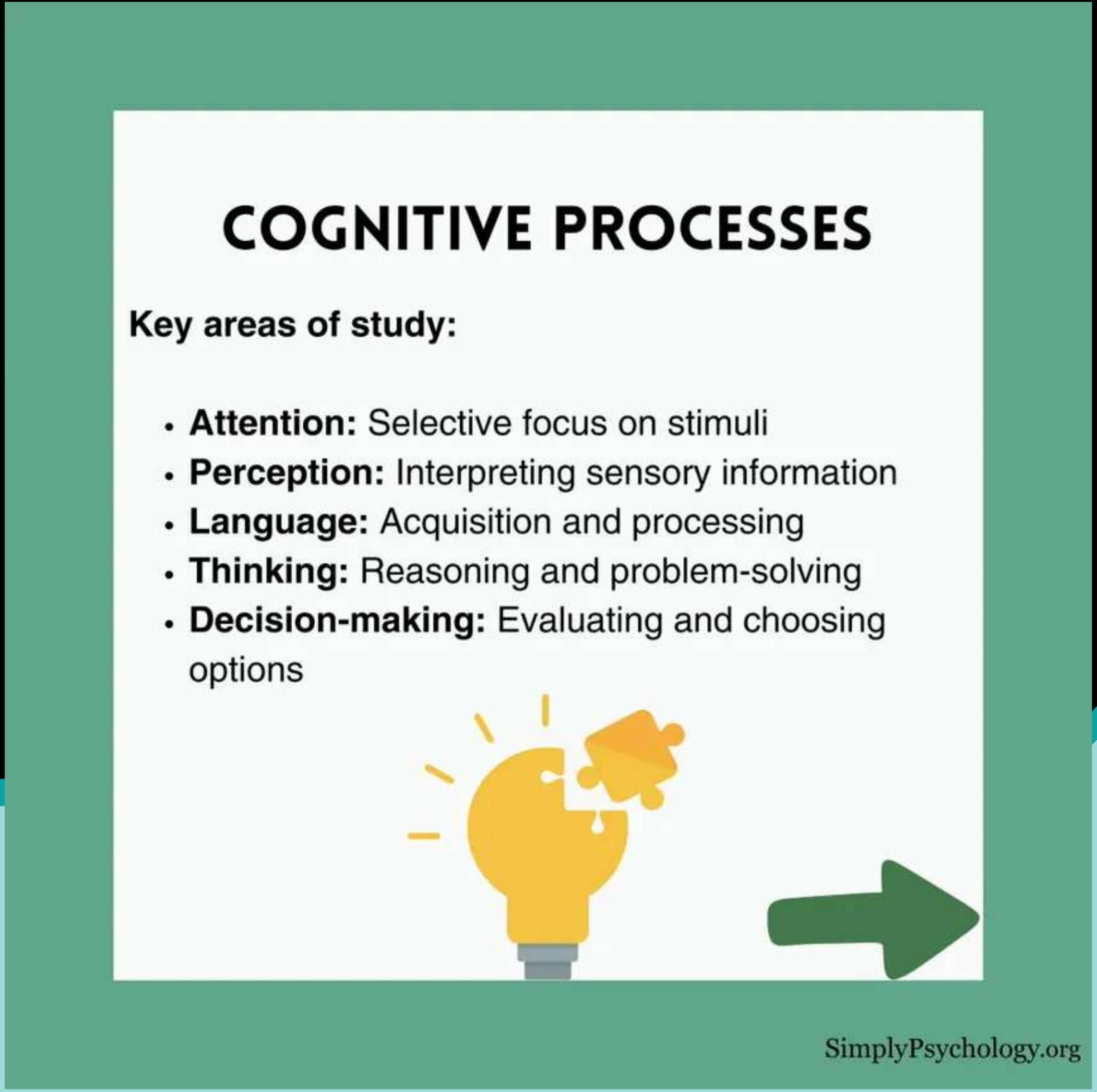
Cognitive Aspects

Objectives

- What is Cognition?
- Types of cognition

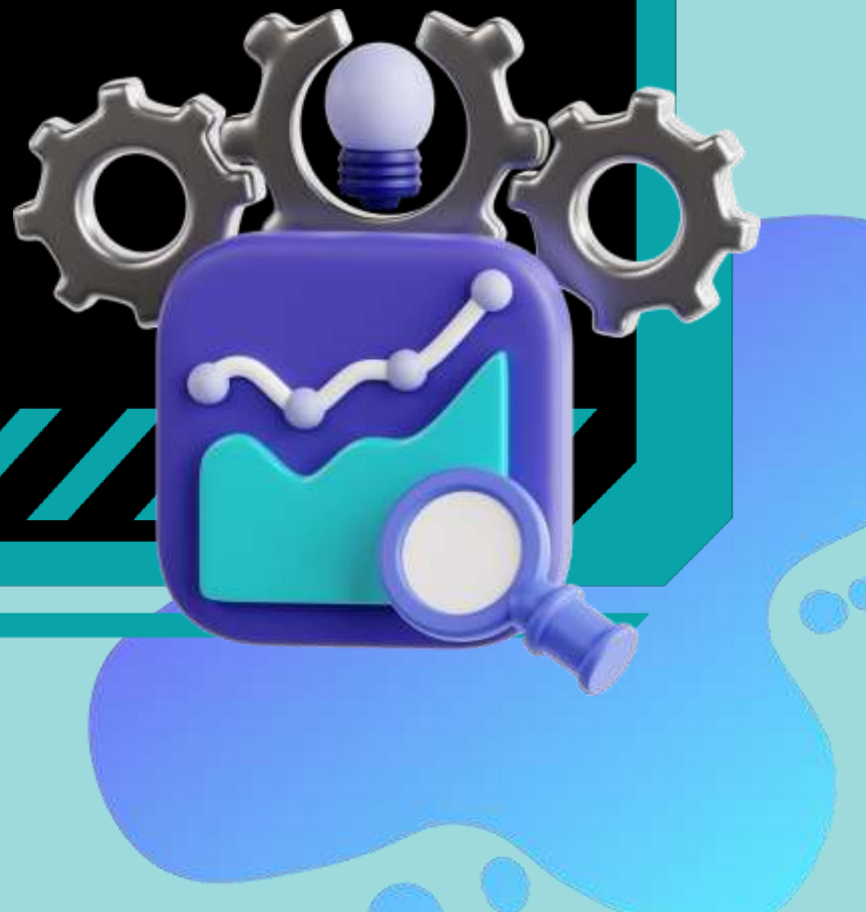
What is Cognition?



What is Cognition?

Types of cognition:

- Thinking
- Remembering
- Learning
- Daydreaming
- Decision Making
- Seeing
- Reading
- Writing
- Talking



2 General Modes of Cognition

1. Experential
2. Reflective

Experential Cognition

It is a state of mind in which we perceive, act, and react to events around us intuitively and effortlessly.

Experential Cognition

Examples include driving a car, reading a book, having a conversation, and playing a video game.

Reflective Cognition

Slow thinking involve mental effort, attention, judment, and decision-making. It involves analyzing past experiences.

This cognition leads to new ideas and creativity.

Cognition kinds of Processes

- Attention
- Perception
- Memory
- Learning
- Reading, speaking, and listening
- Problem solving, planning, reasoning, and decision making

1.Attention

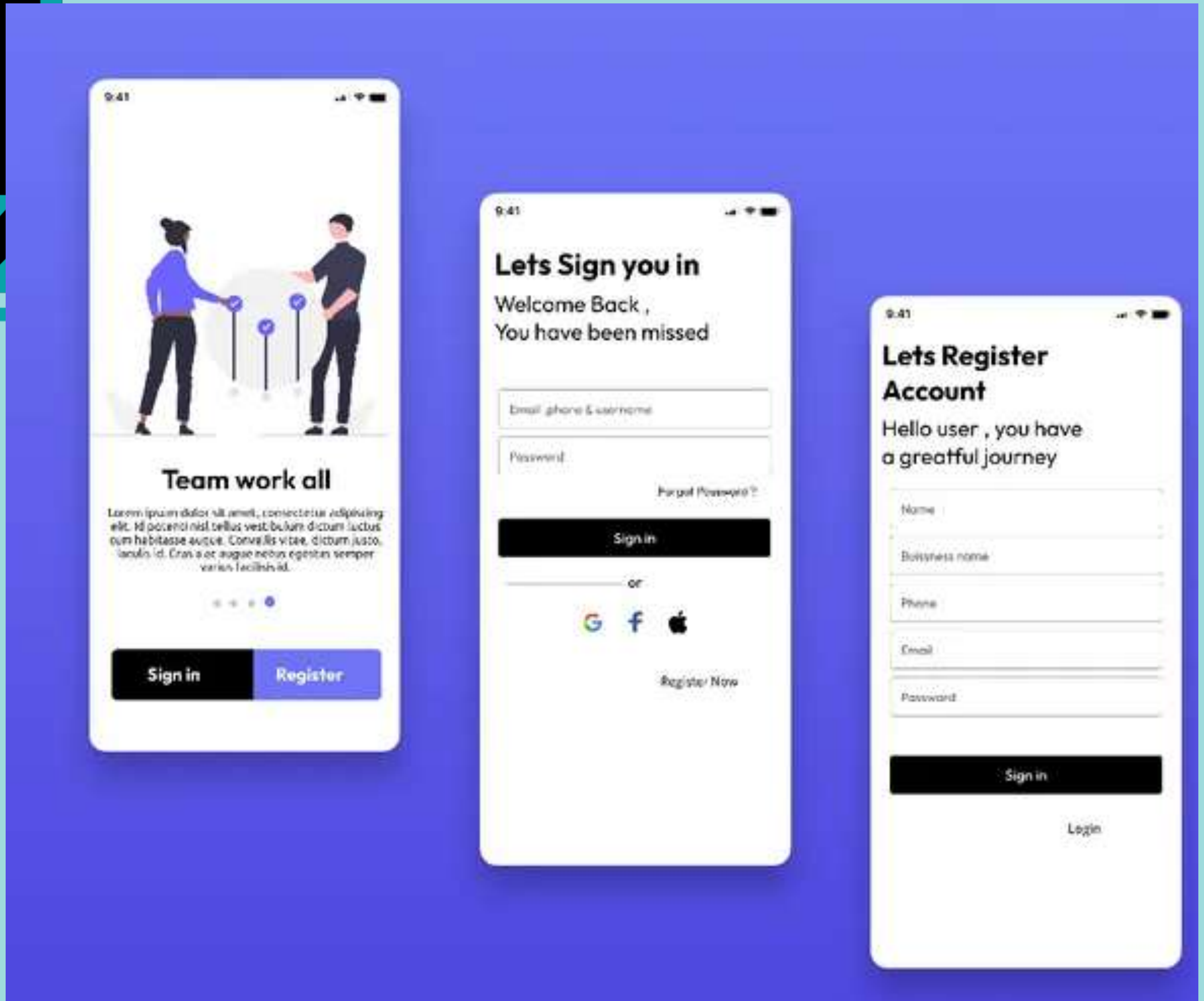
The process of selecting things to concentrate on, at a point in time, from range of possibiities available.

Attention involves our auditory and/or visual senses.

1.Attention (Design Implications)

Use animated grahics, colors, underlining, ordering of items, sequence of information, and spacing of items.

Avoid too much infomation



1.Perception

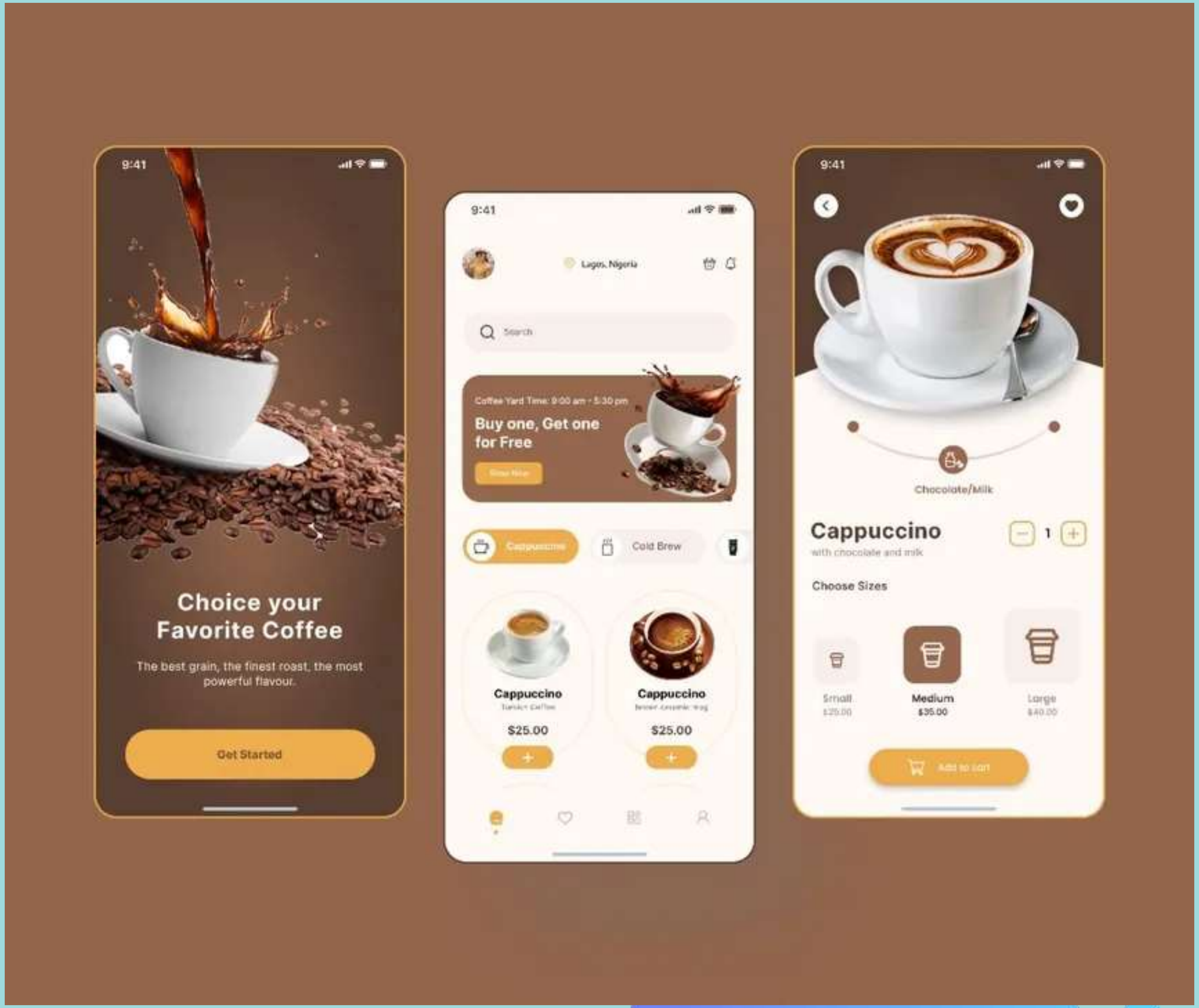
It refers to how information is acquired from the environment via the different sense organs, like our eyes, ears, fingers and transformed into experiences of objects, events sounds, tastes(Roth, 1986)

1.Perception(Design Implications)

Graphical representation enable users to readily distinguish their meaning.

Bordering and spacing are effectie visual ways of grouping information.

Text should be distinguishable from the background.



1.Memory

It involves recalling various kinds of knowledge that allow us to act appropriately.

A filterring process is used to decide what information gets further processed and memorized.

1.Memory

People are much better at recognizing things than recalling things.

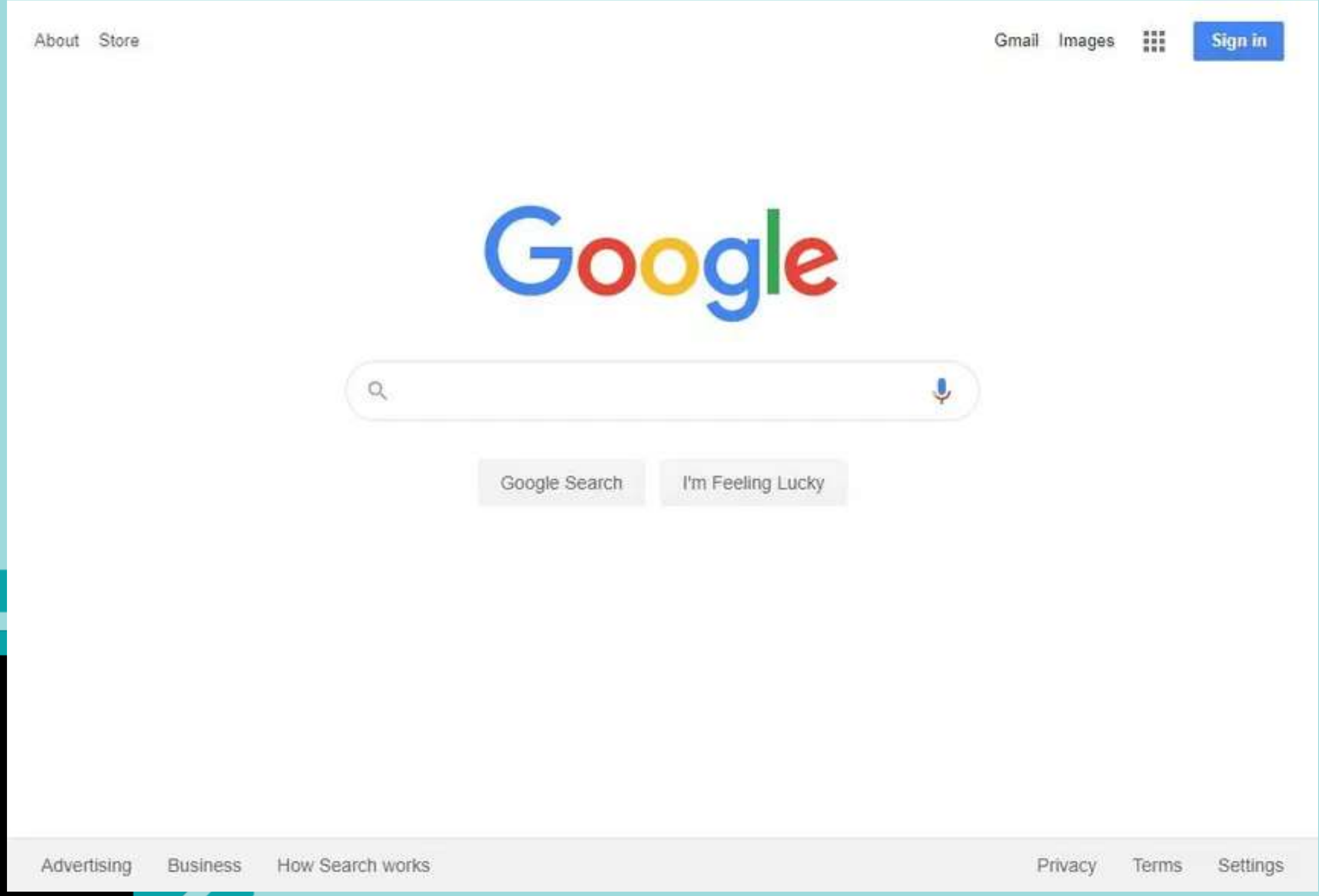
Certain kinds of information are easier to recognize than others.

1.Memory (Design Implications)

Do not overload user's memories with complicated procedures.

Design interfaces that promote recognition by using menus, icons, ans consistenlty placed objects.

Provide users ways of encoding digital information by using categories, color, tagging, icons, etc.



1.Learning

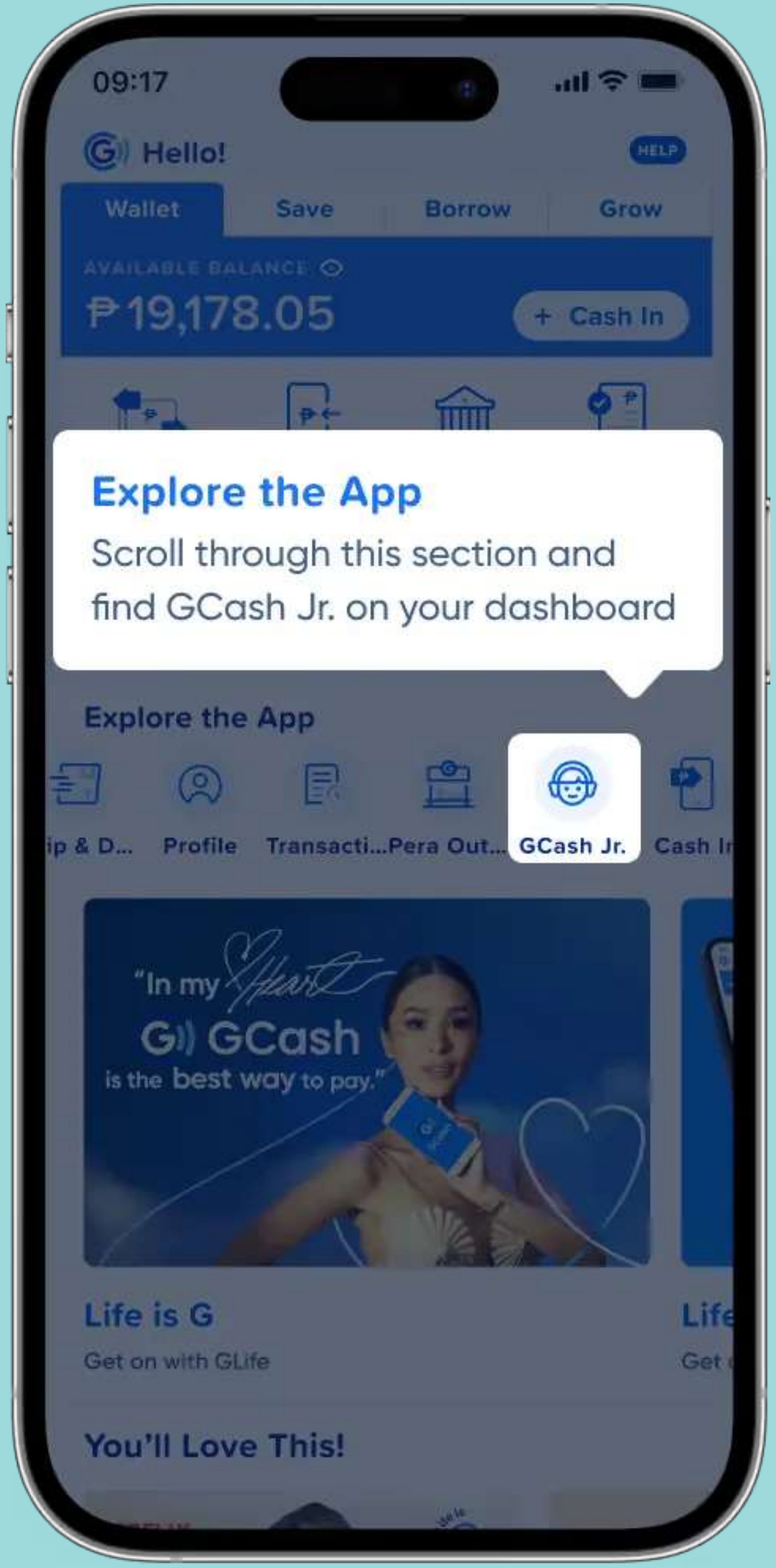
People find it hard to learn by following a set of instructions in manual.

GUIs and direct manipulation interfaces are good environment for supporting active learning.

1.Learning (Design Implications)

Design interfaces that encourages exploration.

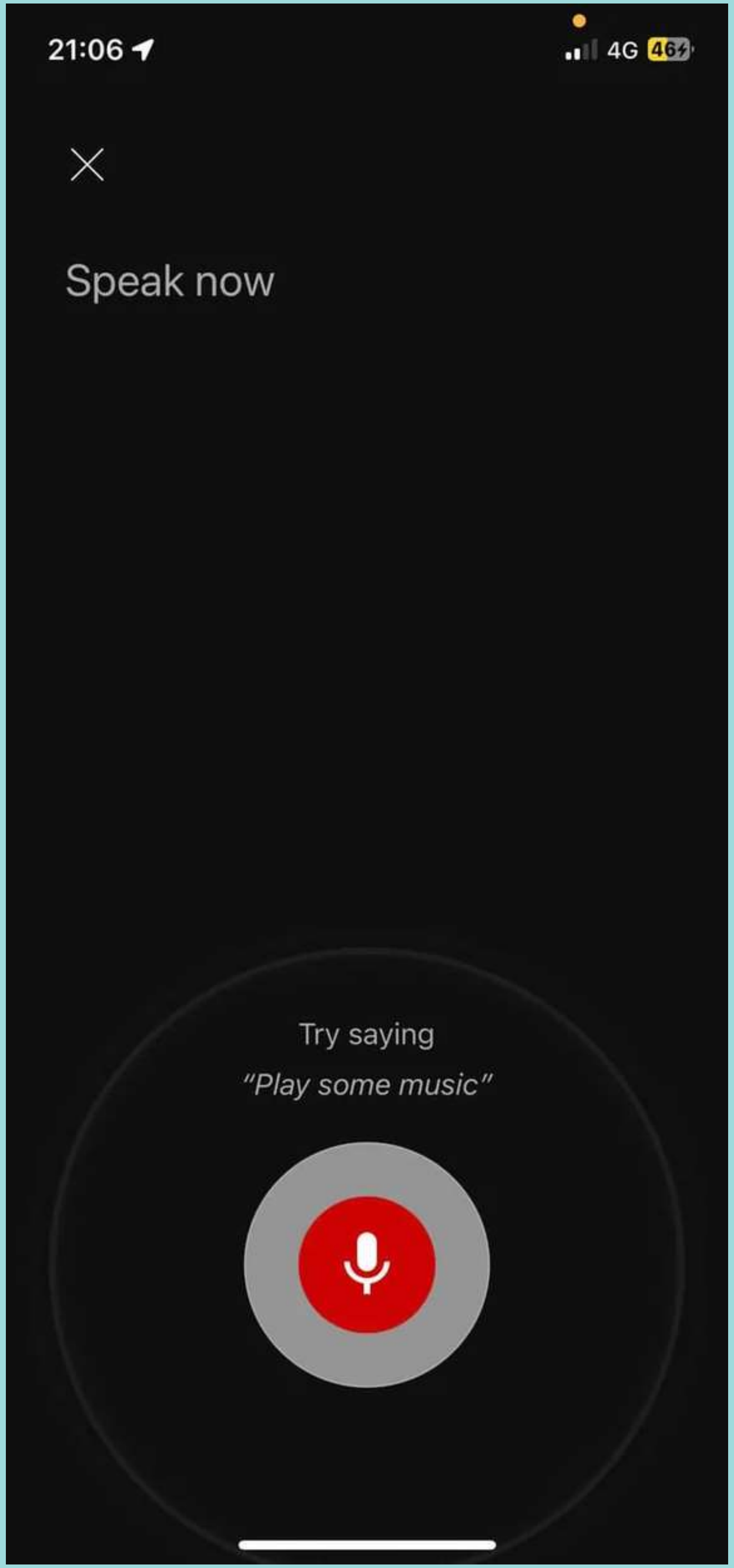
Design interafaces that guide users to select appropriate actions when initially learning.



1.Reading, Speaking, and Listening

Written language is permanent while listening is transient. Reading can be quicker than speaking or listening.

Listening requires less cognitive effort than reading or speaking.



1.Reading, Speaking, and Listening

Keep the length of the speech-based menus ans instruction to a minimum.

Emphasis the intonation of AI speech, voices, as they harder to understand in human voice.

Provide opportunities for making text in screen larger.

1.Problem Solving, Planning, Reasoning, and Decision Making

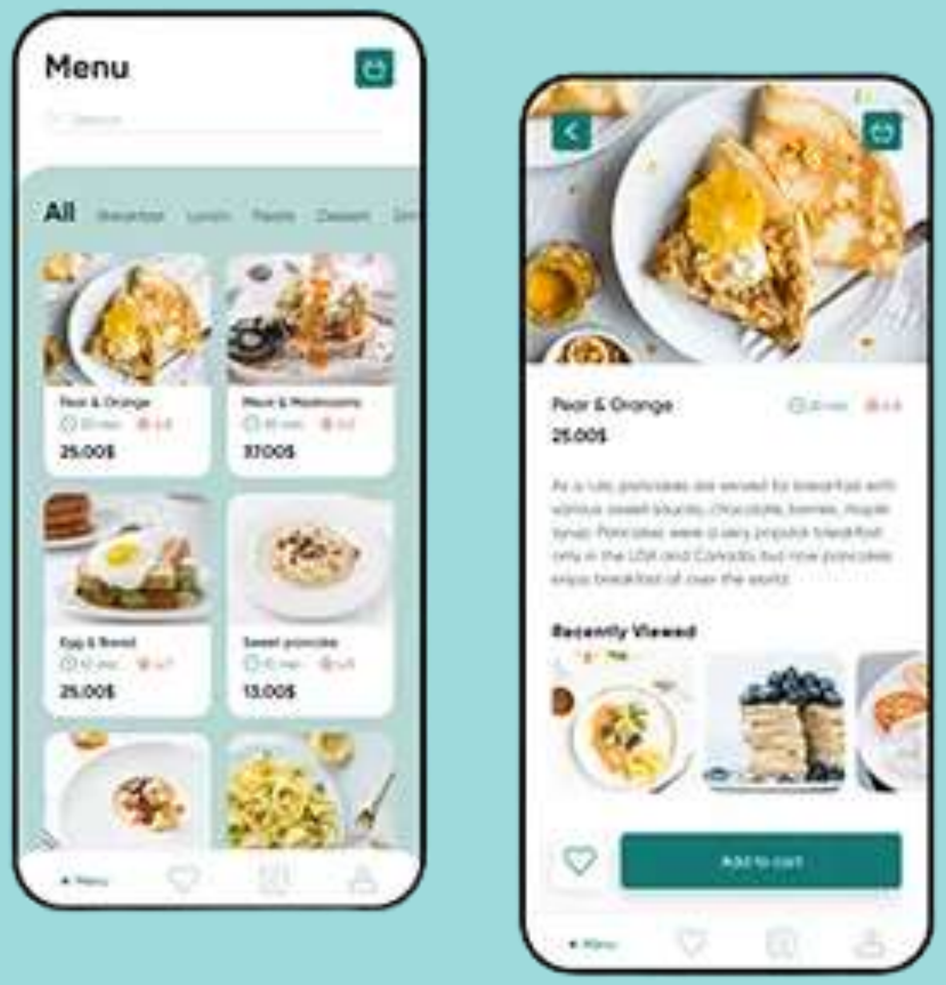
This include thinking about what to do, what are the options are, and what are the consequences.

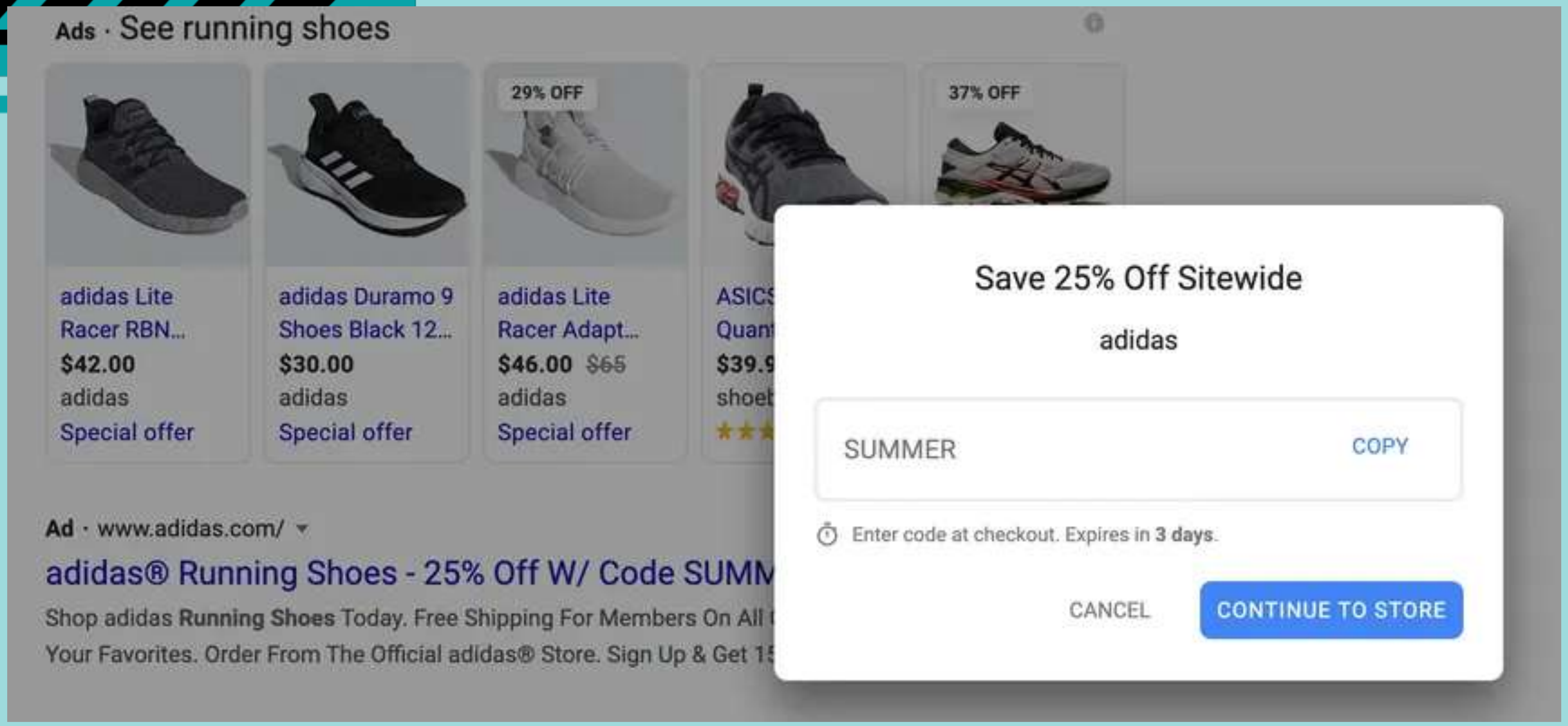
This tells us “How people make decisions when confronted with information overload, such as when shopping on the web or at the store.”

1.Problem Solving, Planning, Reasoning, and Decision Making

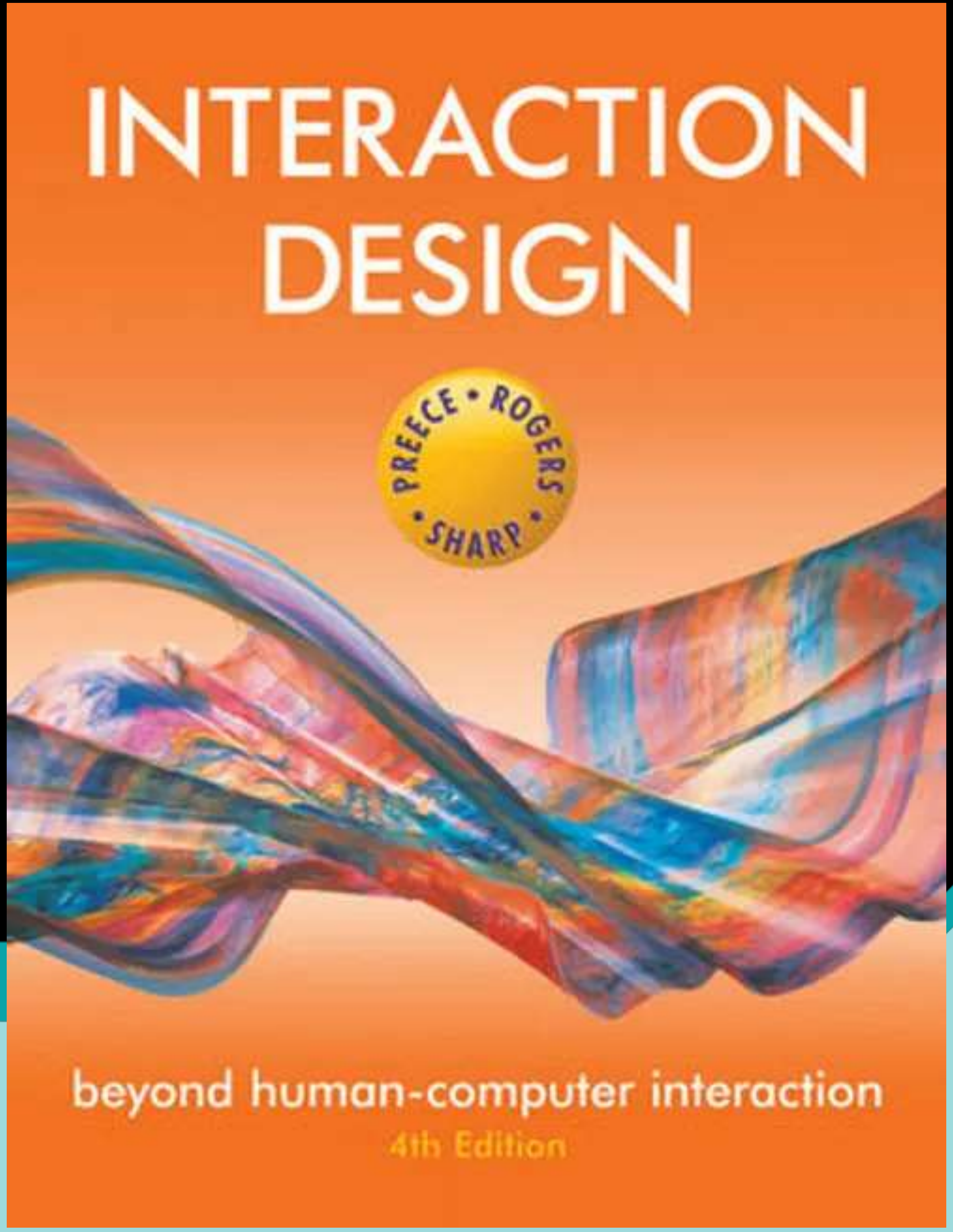
Provide additional information that is easy to access for users to understand more about the product.

Use simple and memorable functions that support decision making and planning.





Reference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